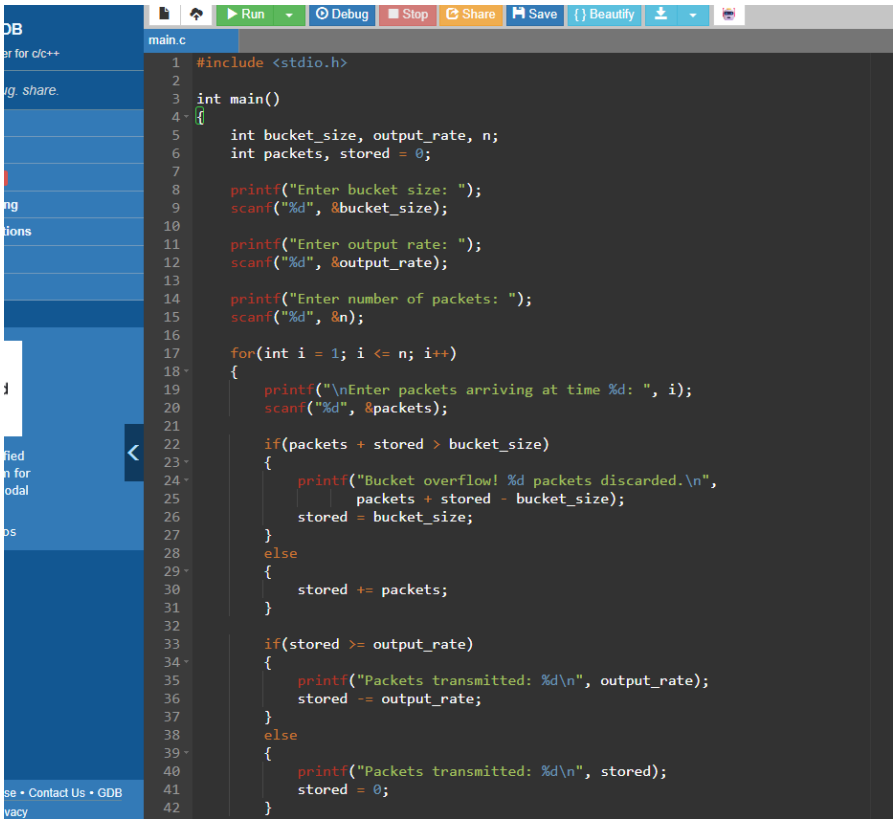
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Batch : B2




```c
#include <stdio.h>

int main()
{
    int bucket_size, output_rate, n;
    int packets, stored = 0;

    printf("Enter bucket size: ");
    scanf("%d", &bucket_size);

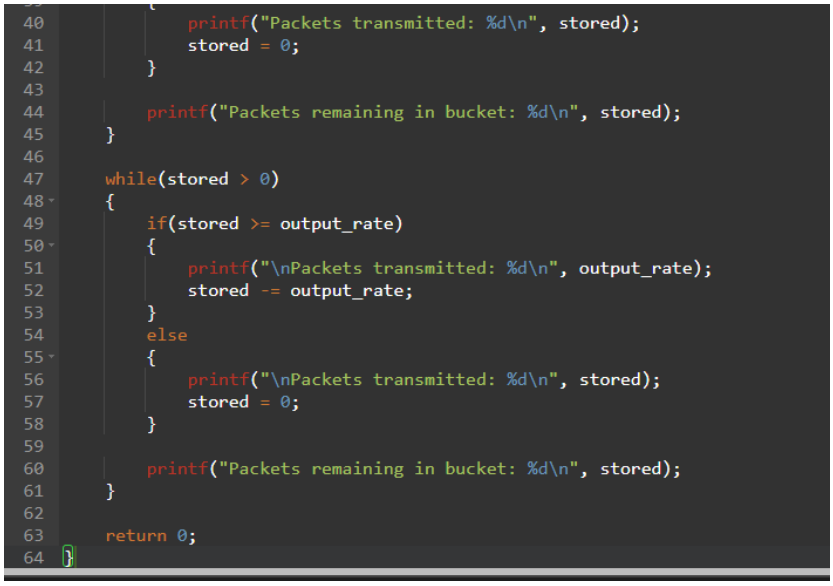
    printf("Enter output rate: ");
    scanf("%d", &output_rate);

    printf("Enter number of packets: ");
    scanf("%d", &n);

    for(int i = 1; i <= n; i++)
    {
        printf("\nEnter packets arriving at time %d: ", i);
        scanf("%d", &packets);

        if(packets + stored > bucket_size)
        {
            printf("Bucket overflow! %d packets discarded.\n",
                   packets + stored - bucket_size);
            stored = bucket_size;
        }
        else
        {
            stored += packets;
        }

        if(stored >= output_rate)
        {
            printf("Packets transmitted: %d\n", output_rate);
            stored -= output_rate;
        }
        else
        {
            printf("Packets transmitted: %d\n", stored);
            stored = 0;
        }
```



```c
            printf("Packets transmitted: %d\n", stored);
            stored = 0;
        }

        printf("Packets remaining in bucket: %d\n", stored);
    }

    while(stored > 0)
    {
        if(stored >= output_rate)
        {
            printf("\nPackets transmitted: %d\n", output_rate);
            stored -= output_rate;
        }
        else
        {
            printf("\nPackets transmitted: %d\n", stored);
            stored = 0;
        }

        printf("Packets remaining in bucket: %d\n", stored);
    }

    return 0;
}
```

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